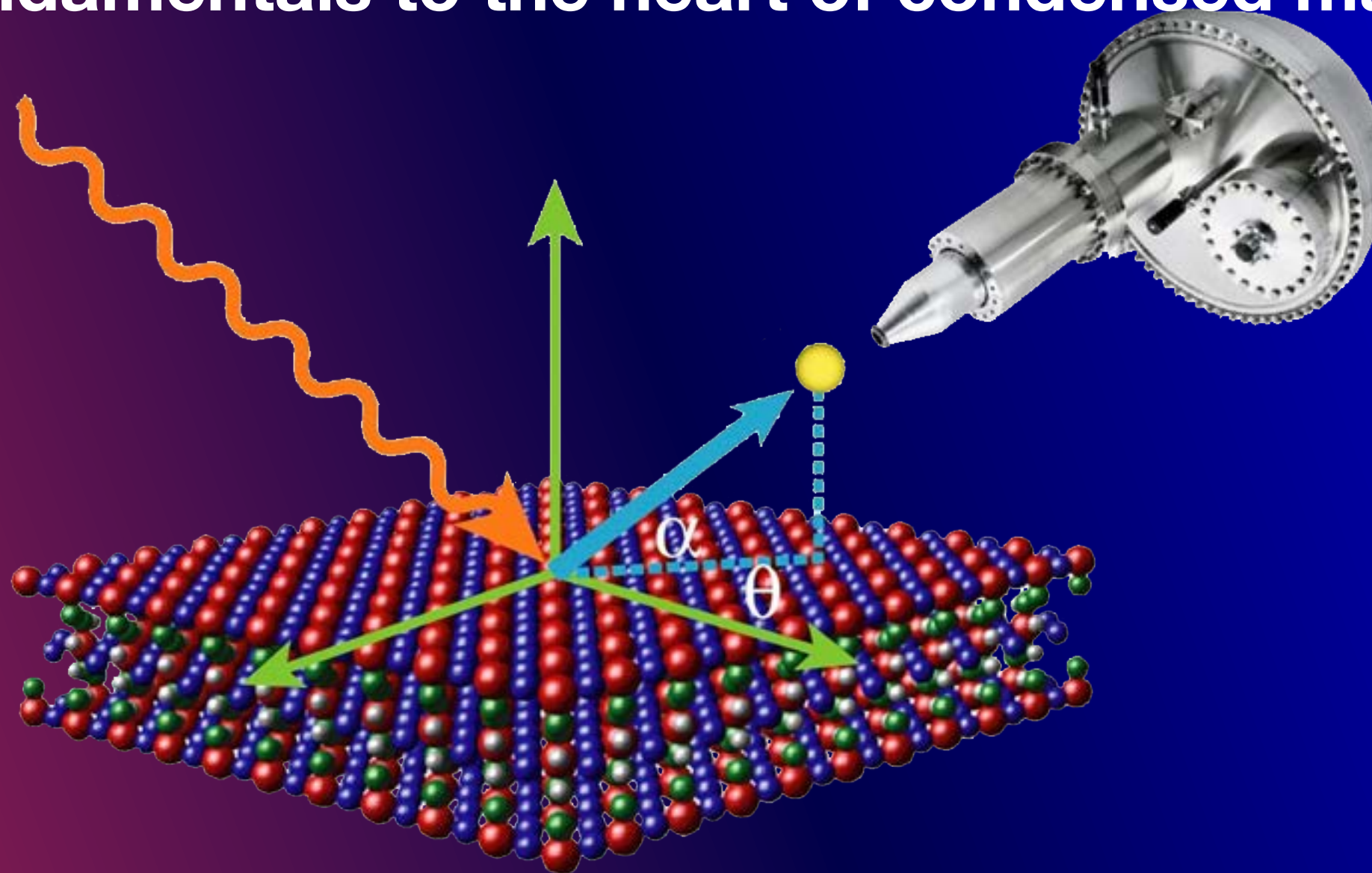


Angle Resolved Photoemission Spectroscopy

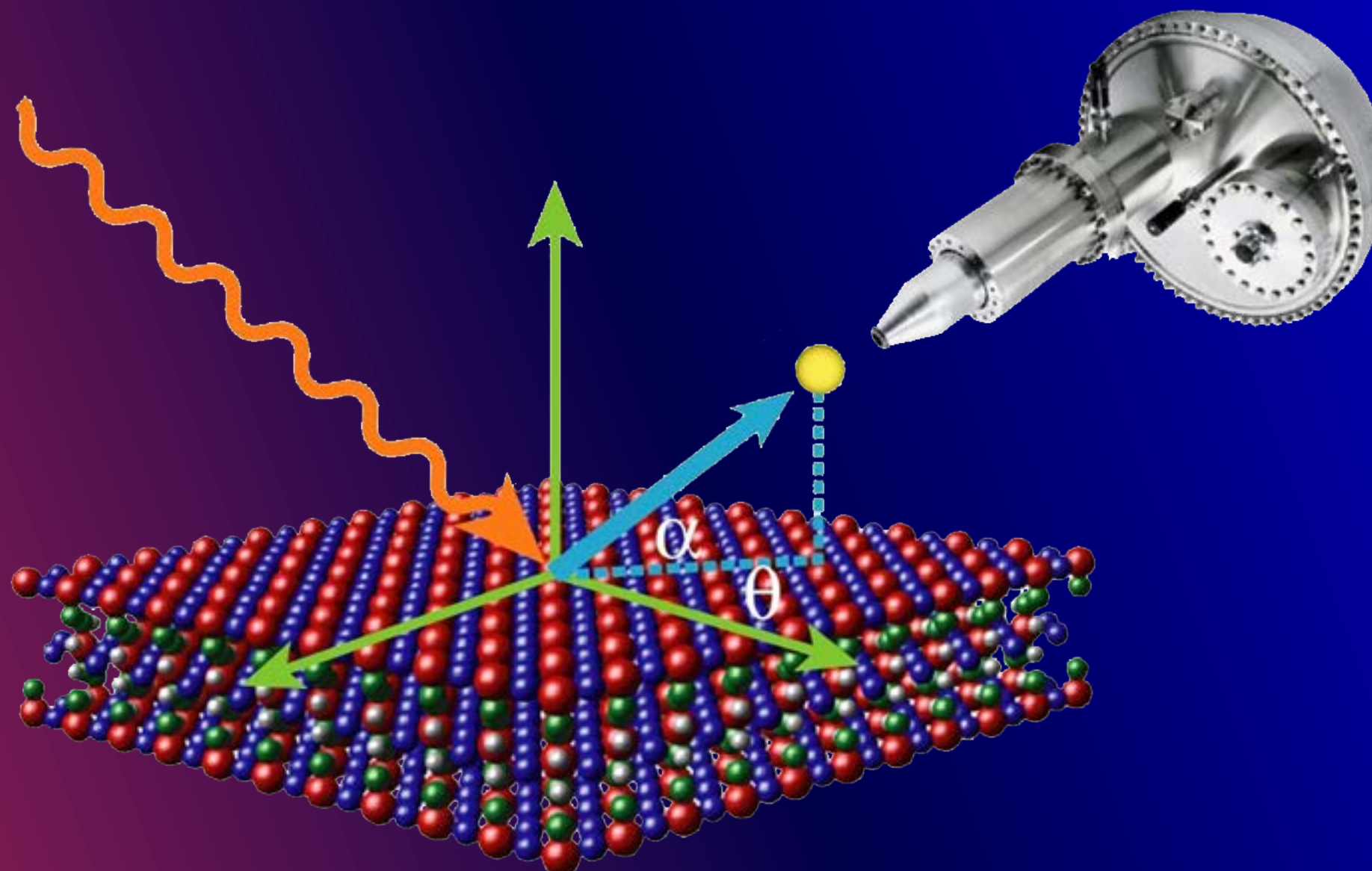
From fundamentals to the heart of condensed matter



6-7 FEBRERO, 2023

Lecture #4 : Frontiers in ARPES

Time, Space, and Spin

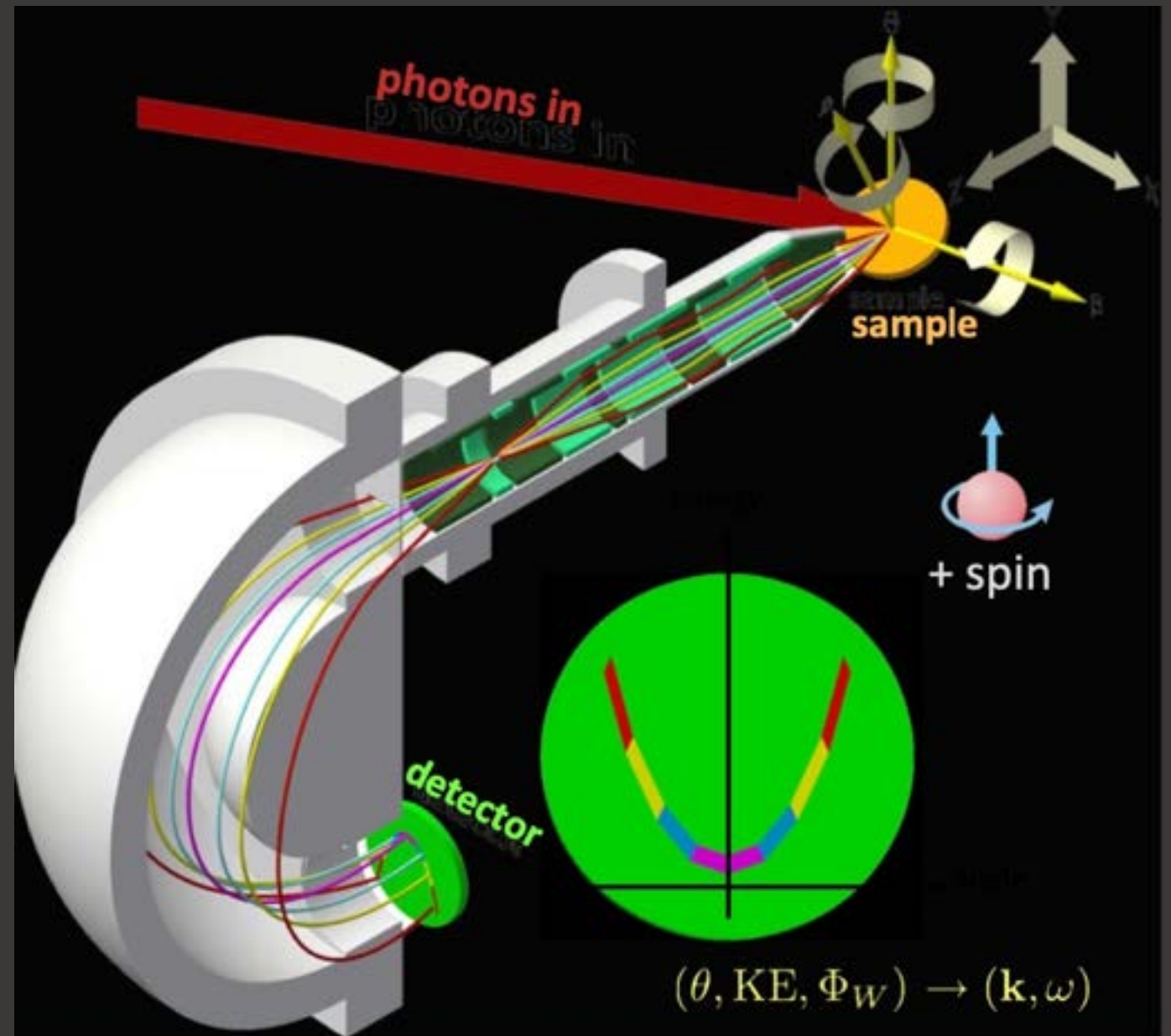


Thanks to Luca Moreschini & Shuolong Yang

6-7 FEBRERO, 2023

ARPES + something else!

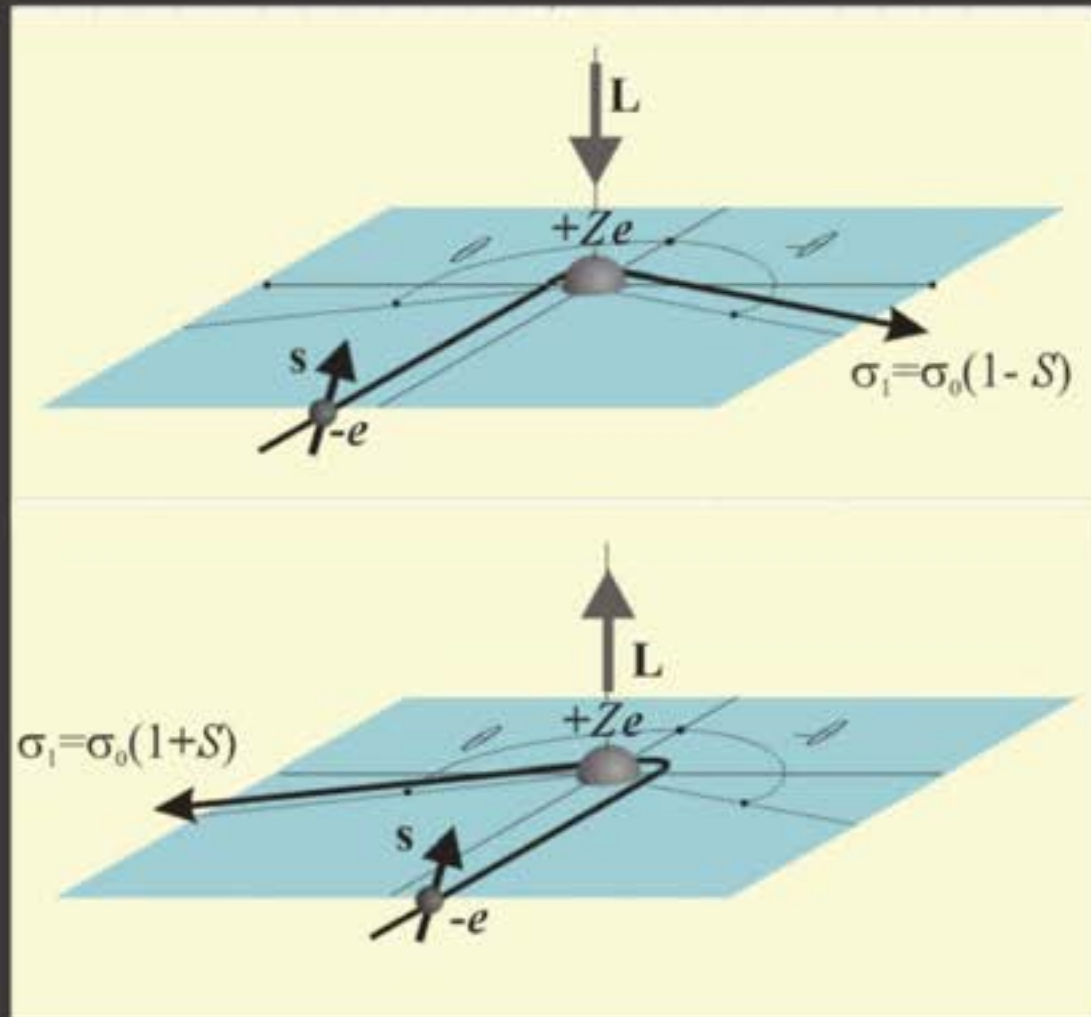
1. Spin detection
2. Time-resolution
3. Spatial resolution



Spin polarimetry



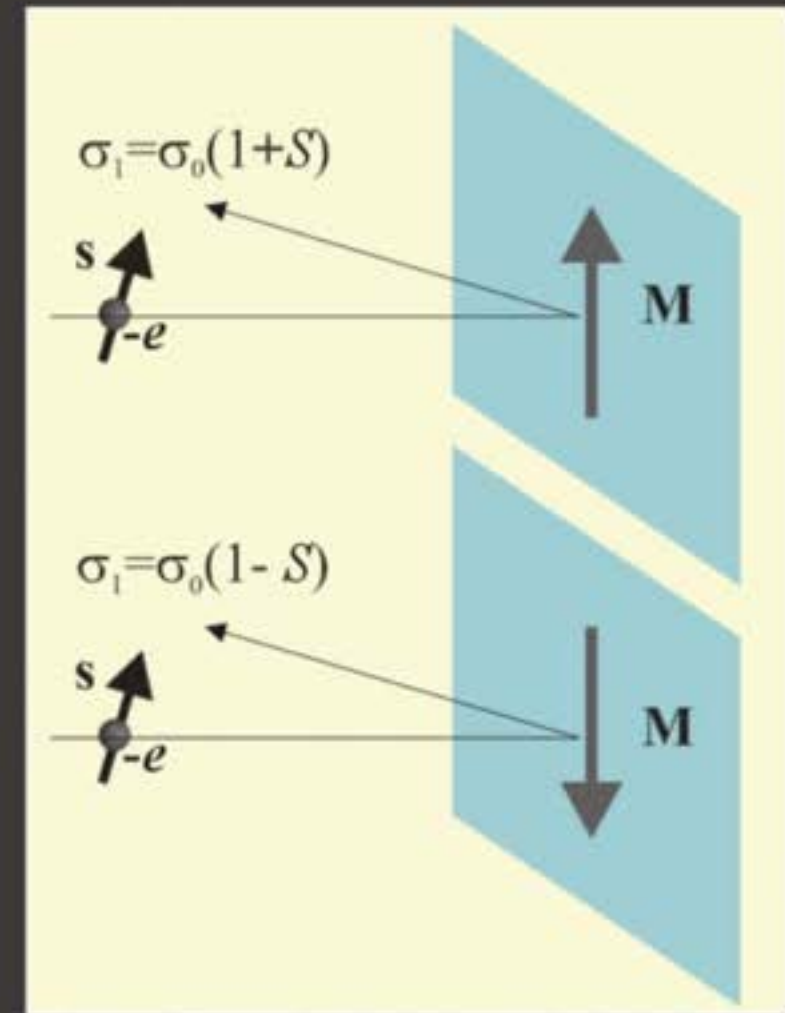
Mott scattering
+ stability, - efficiency



coupling between the atomic orbital
momentum and the spin of the electron

different cross section
for two different scattering directions

exchange scattering (VLEED)
- stability, + efficiency



coupling between the ferromagnet magnetic
moment and the spin of the electron

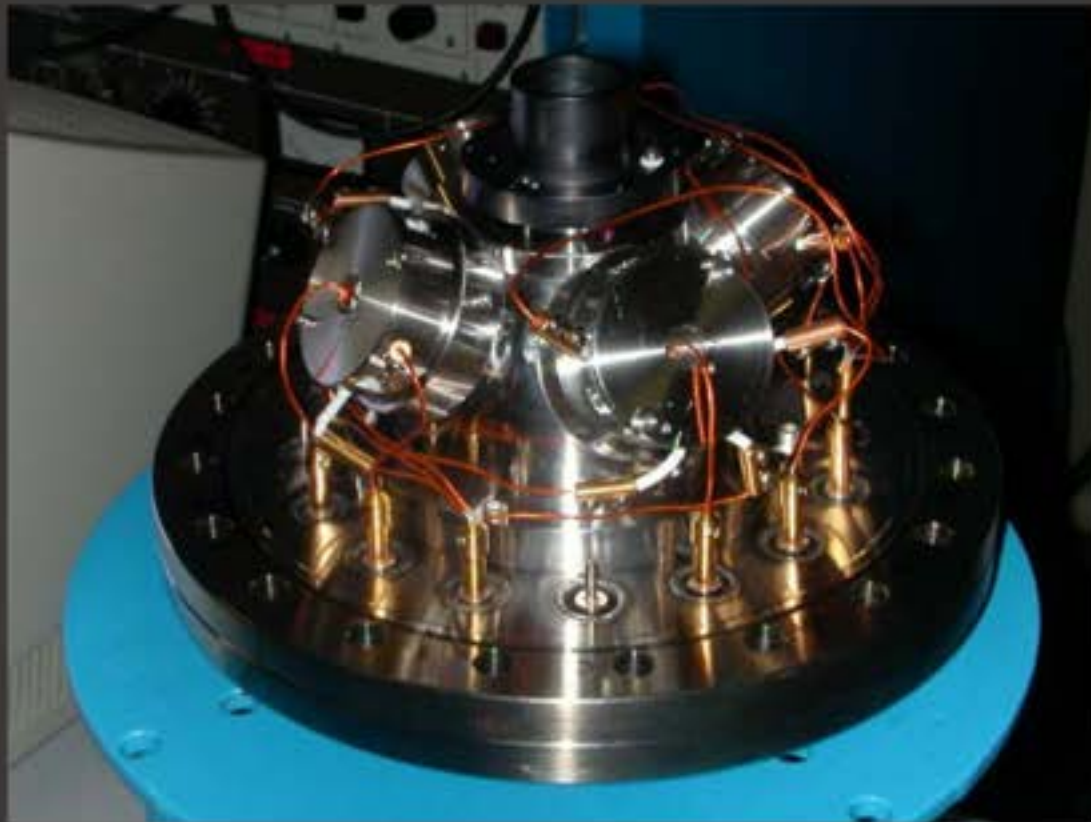
different reflectivity
for two opposite magnetization directions

Spin polarimetry



Mott scattering
+ stability, - efficiency

exchange scattering (VLEED)
- stability, + efficiency



**different cross section
for two different scattering directions**



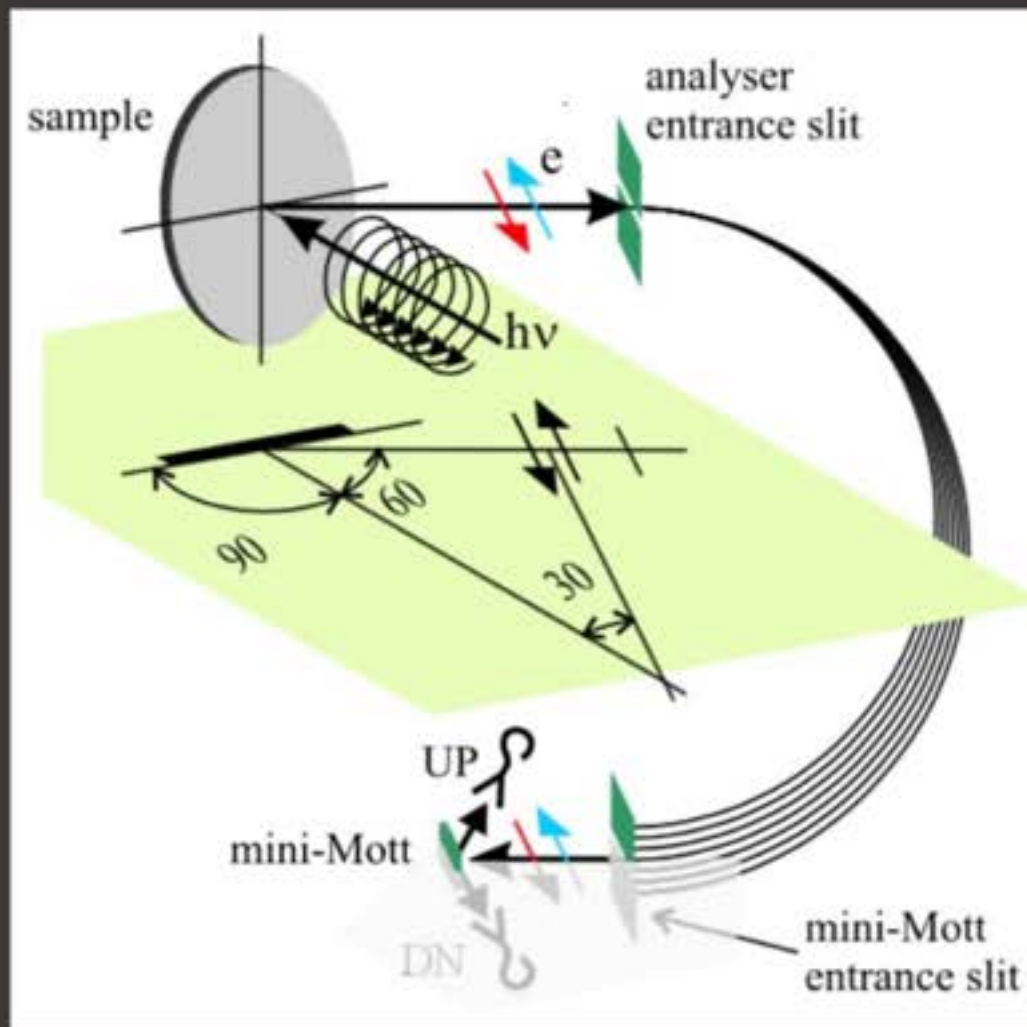
**different reflectivity
for two opposite magnetization directions**

Electron analyzers for spin polarimetry

From hemispherical analyzers to time-of-flight analyzers

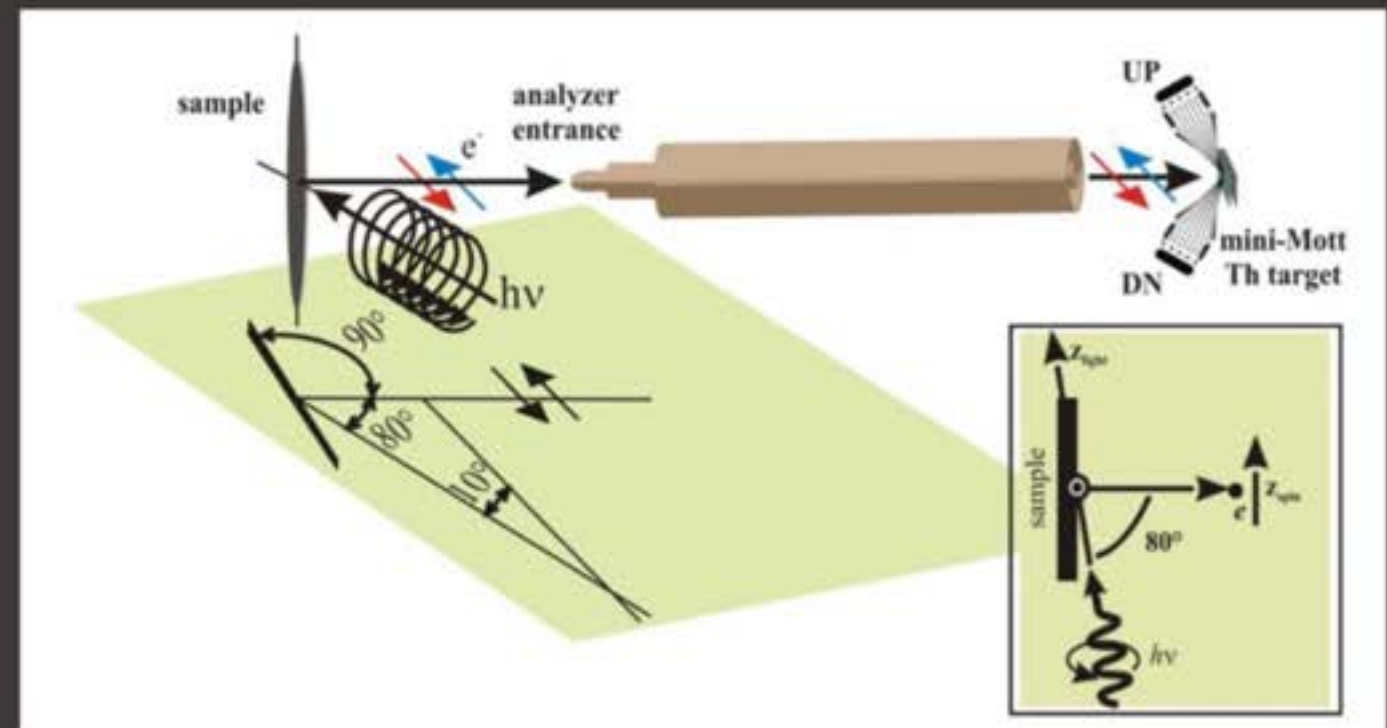
Hemispherical EA-Mott

- + resolution/stability vs $h\nu$
- efficiency (serial acquisition)



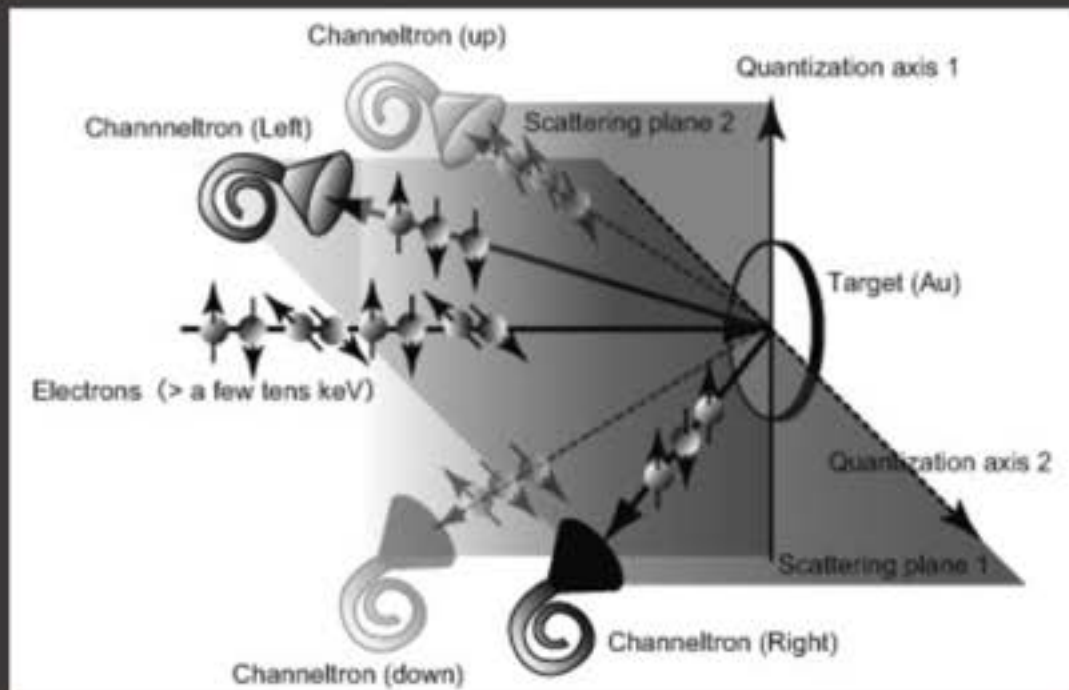
TOF-Mott

- resolution/stability vs $h\nu$
- + efficiency (parallel acquisition)

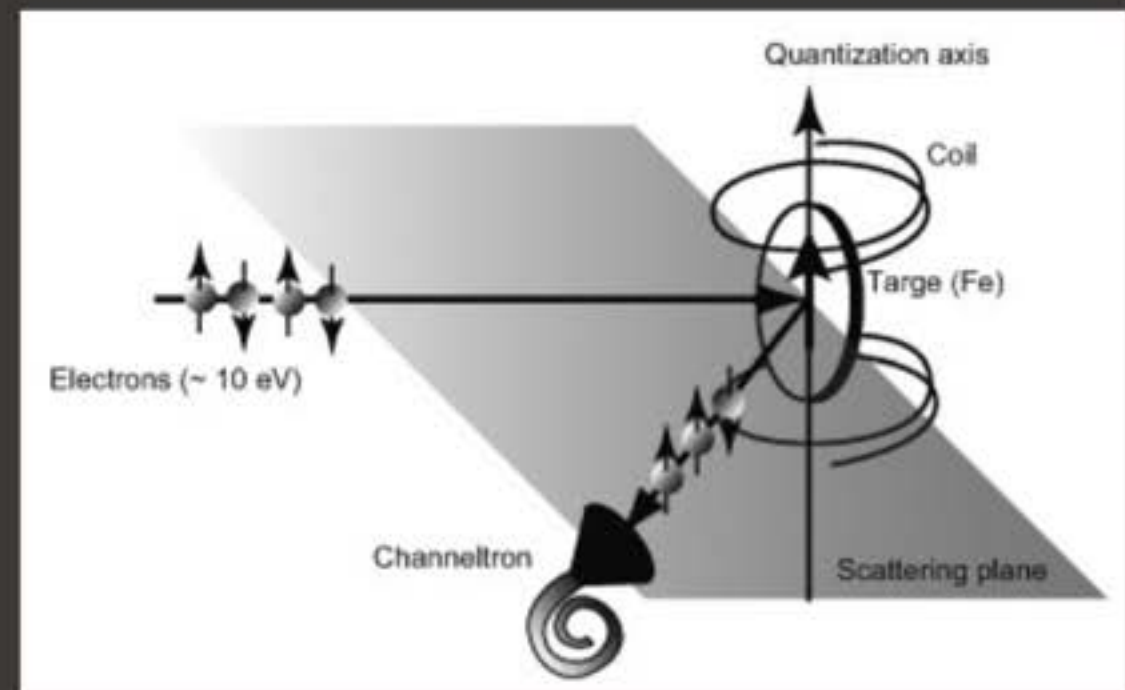


Coupling spin detection and ARPES

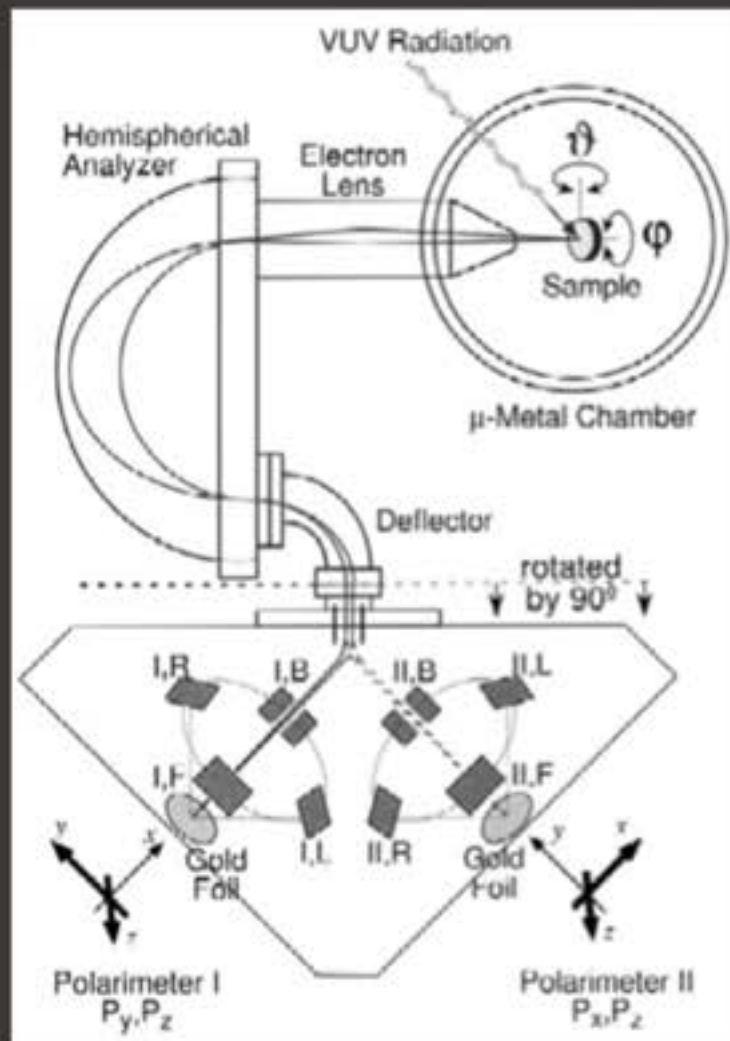
Mott polarimeters
two axes in parallel



VLEED polarimeters
one axis at a time

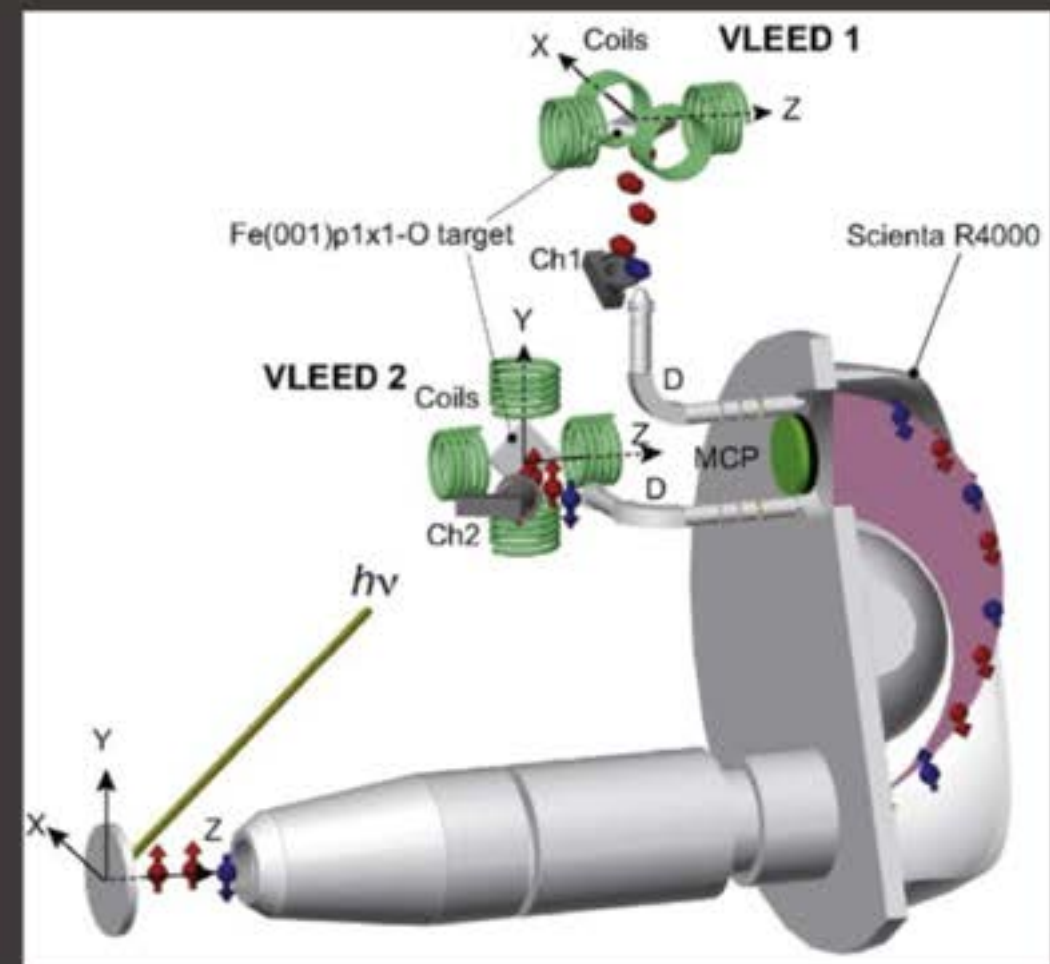


Coupling spin detection and ARPES - a few examples



Hemispherical EA + Mott

M. Hoesch *et al.*, *JESRP* 124, 263 (2002)



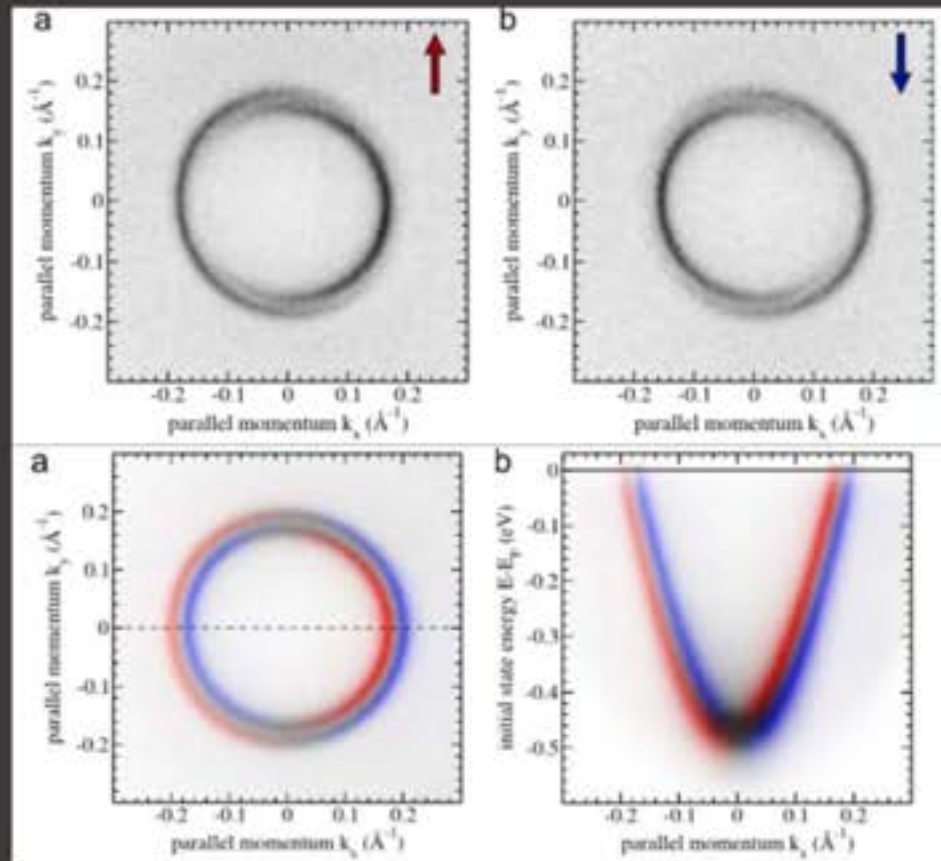
Hemispherical EA + VLEED

T. Okuda *et al.*, *Rev. Sci. Instrum.* 82, 103302 (2011)

T. Okuda *et al.*, *Rev. Sci. Instrum.* 201, 23 (2015)

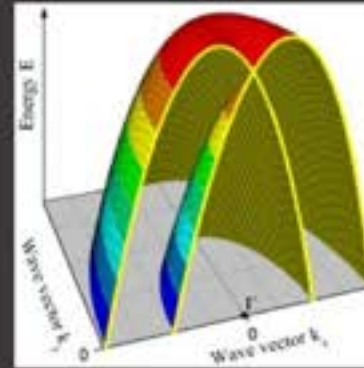
Rashba systems - examples

Au(111)

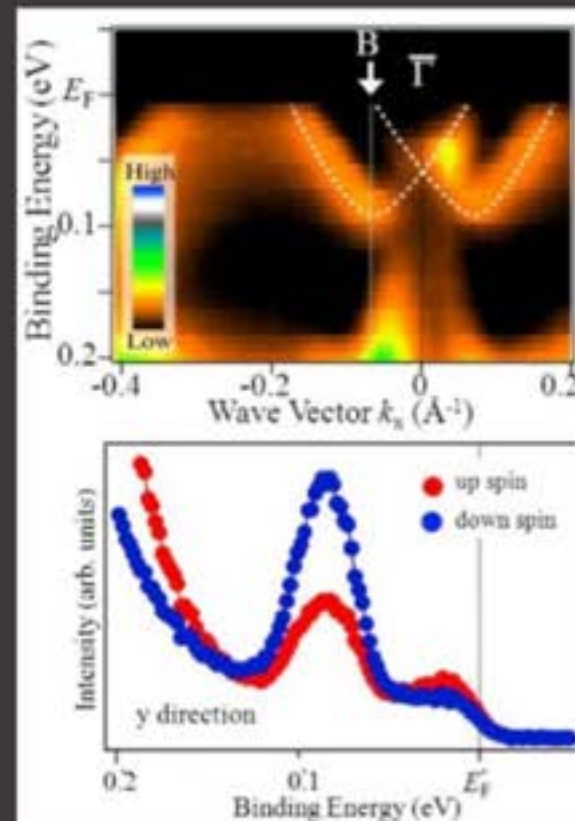


Shockley states in noble metals

C. Tusche *et al.*, Ultramicroscopy, **159**, 520 (2015)



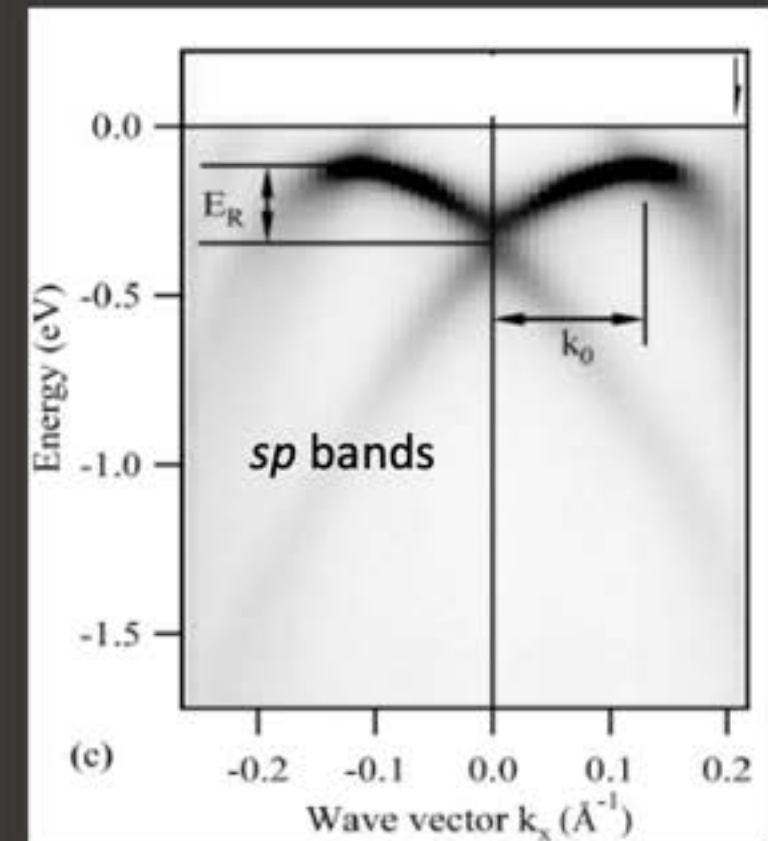
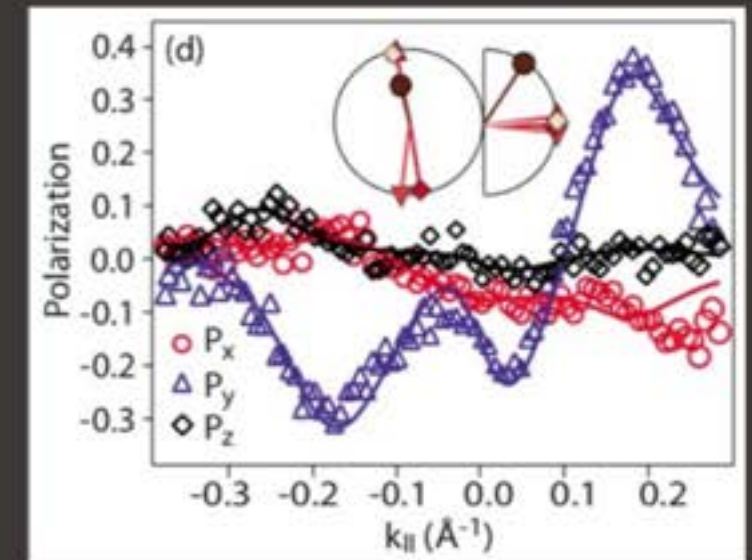
Bi (111)



surface states in heavy metals

A Takayama *et al.*,
New J. Phys. **16** 055004 (2014)

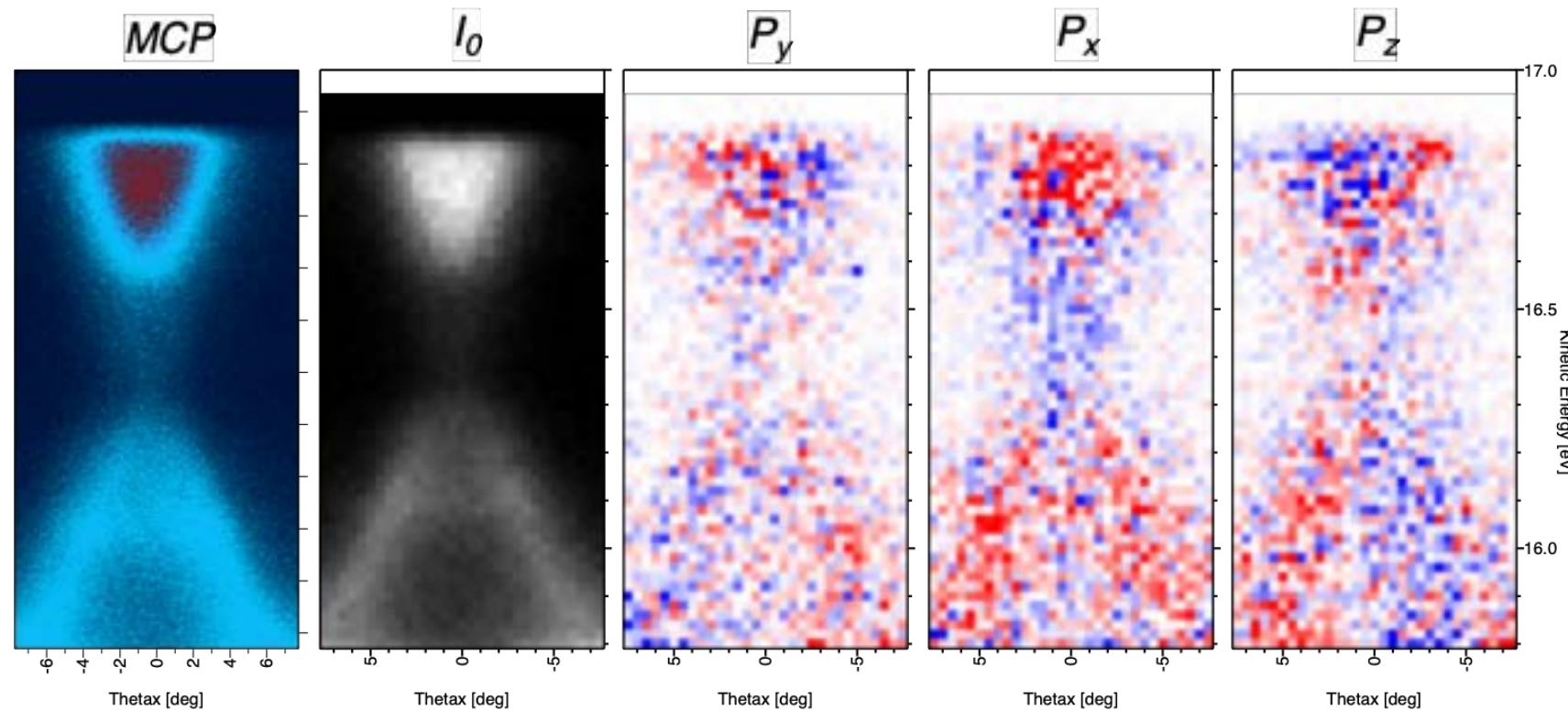
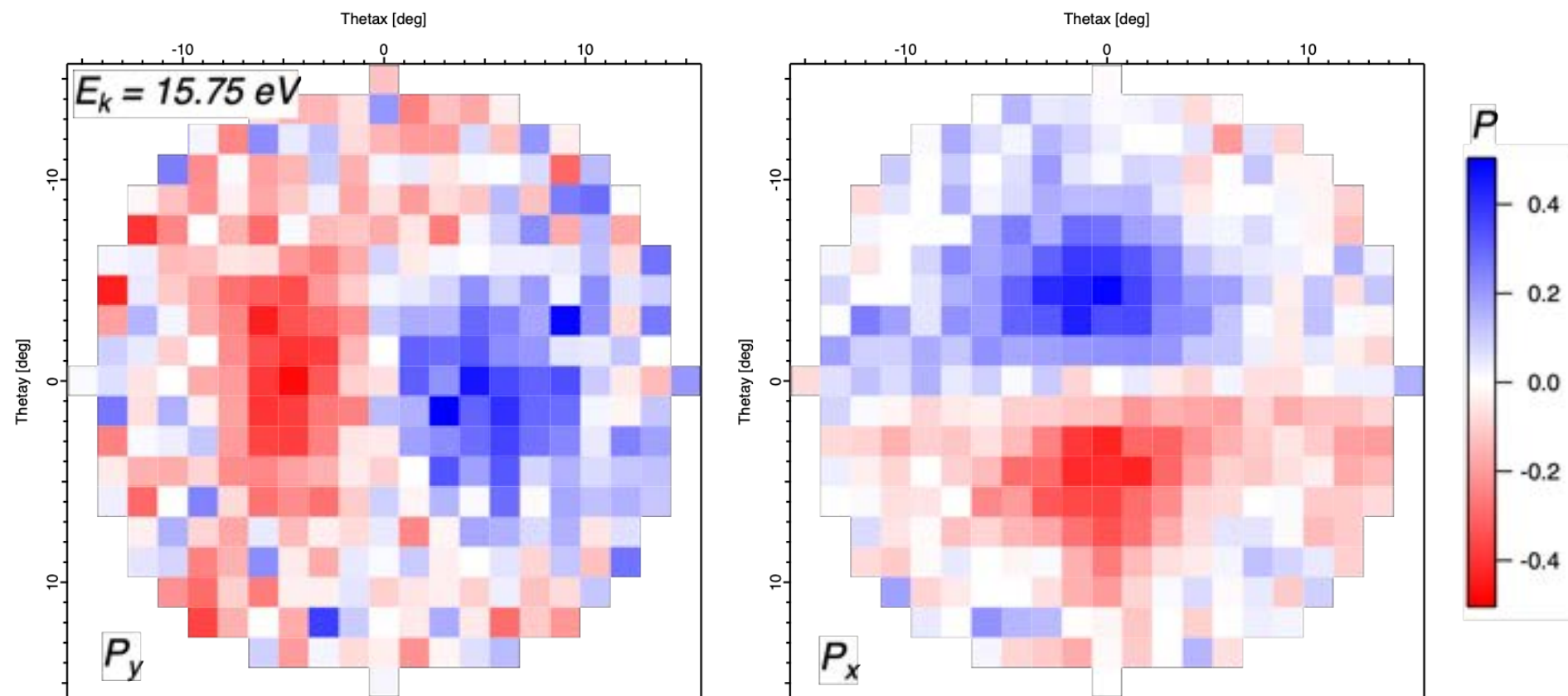
Bi/Ag(111)



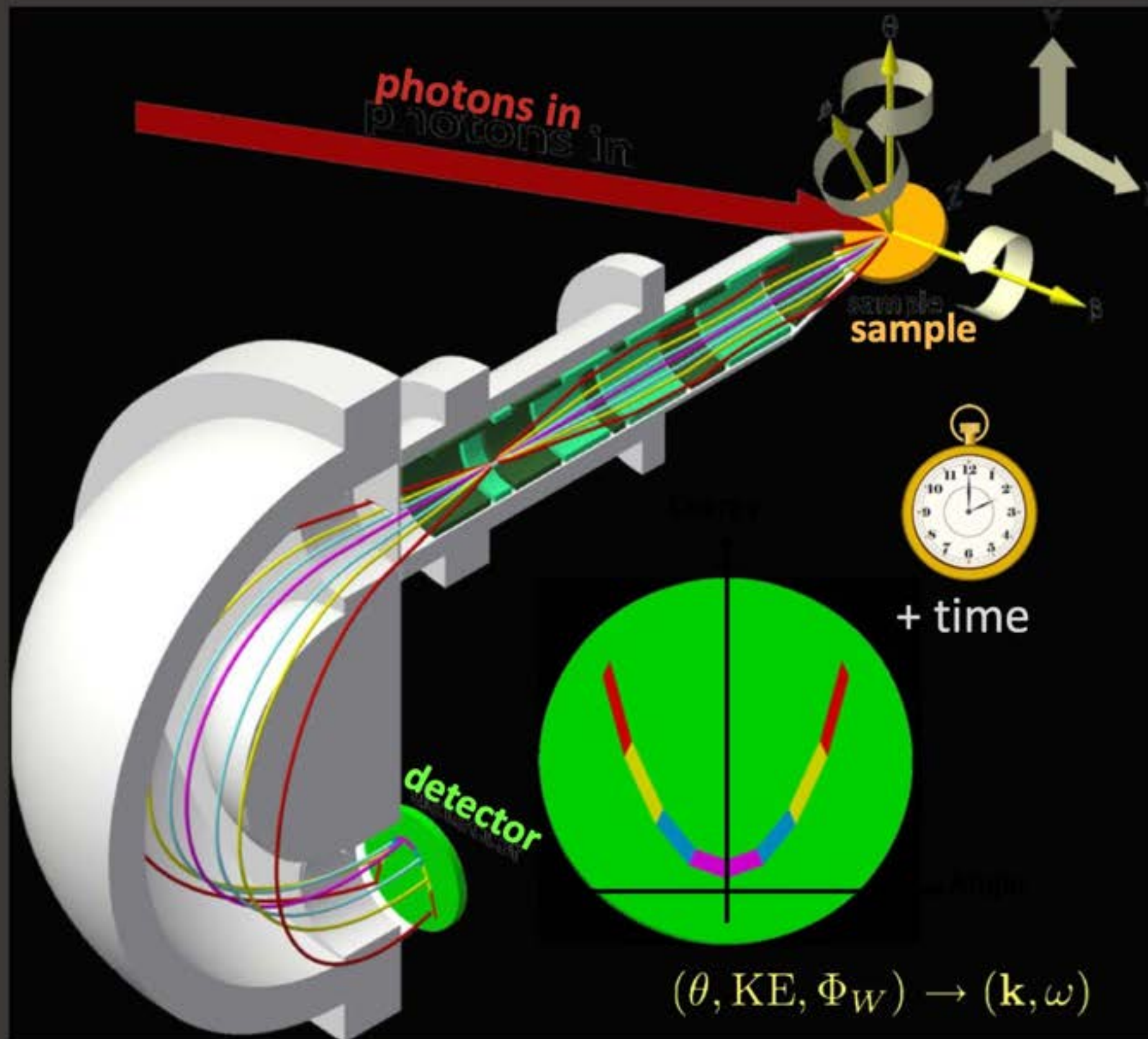
interface states in surface alloys

F. Meier *et al.*, Phys. Rev. B **77**, 165431 (2008)

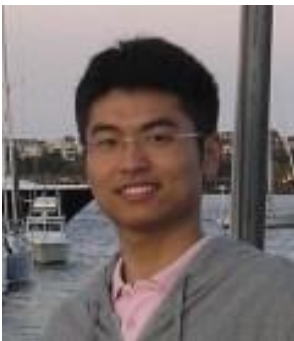
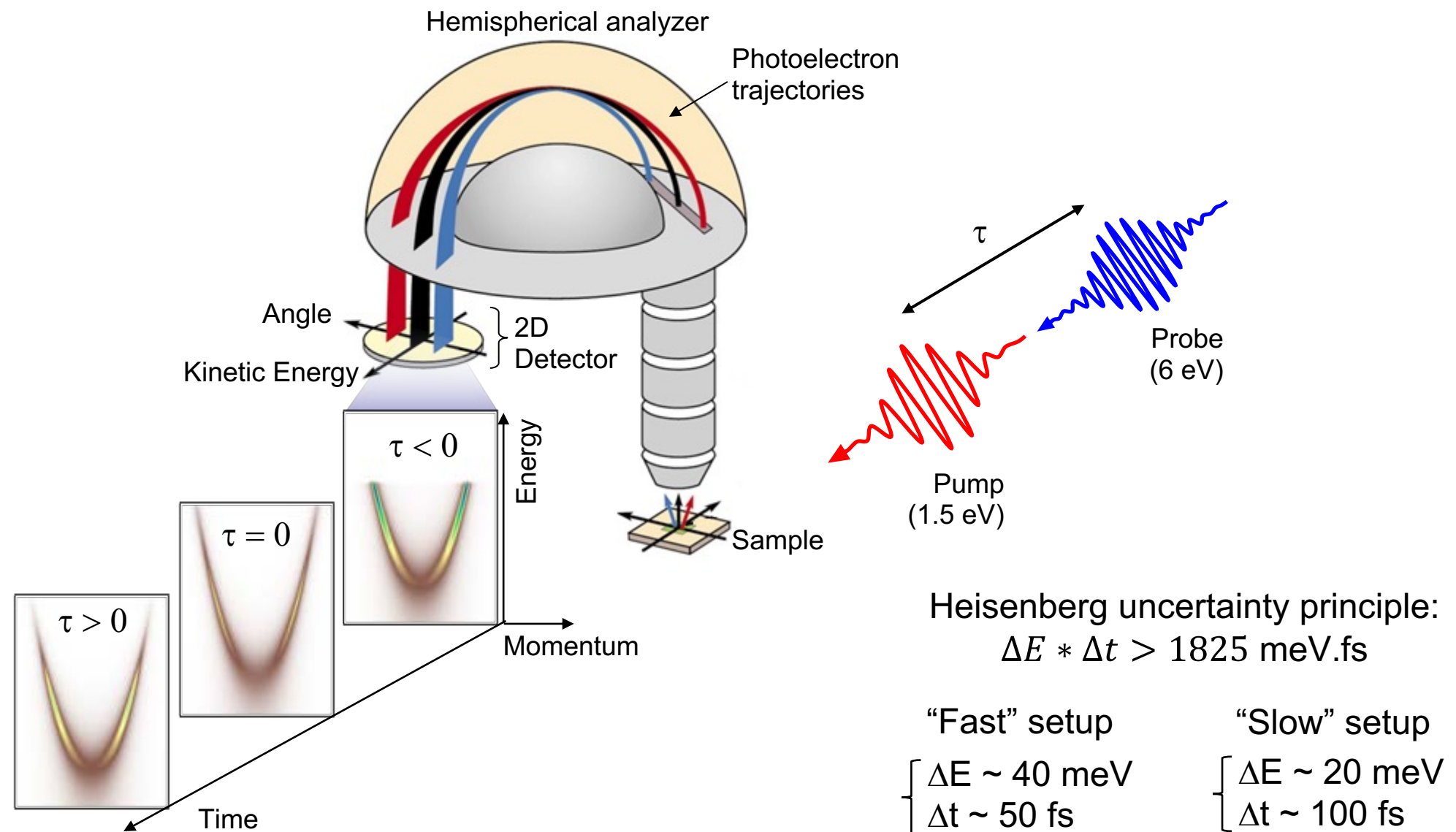
Spin-Resolved ARPES using VLEED detection



Angle-resolved photoelectron spectroscopy + something else



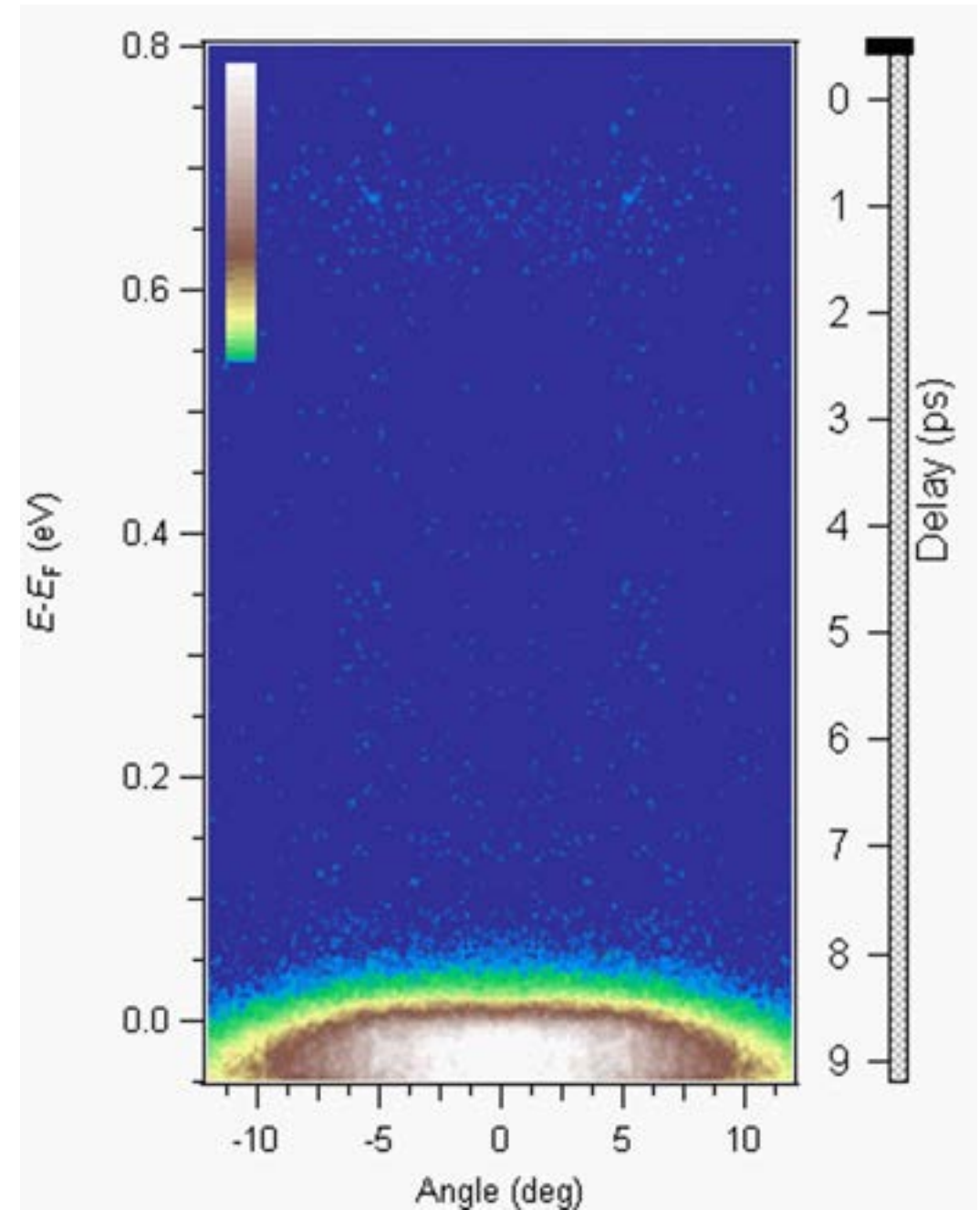
Time-Resolved ARPES : Visualizing Electron Dynamics



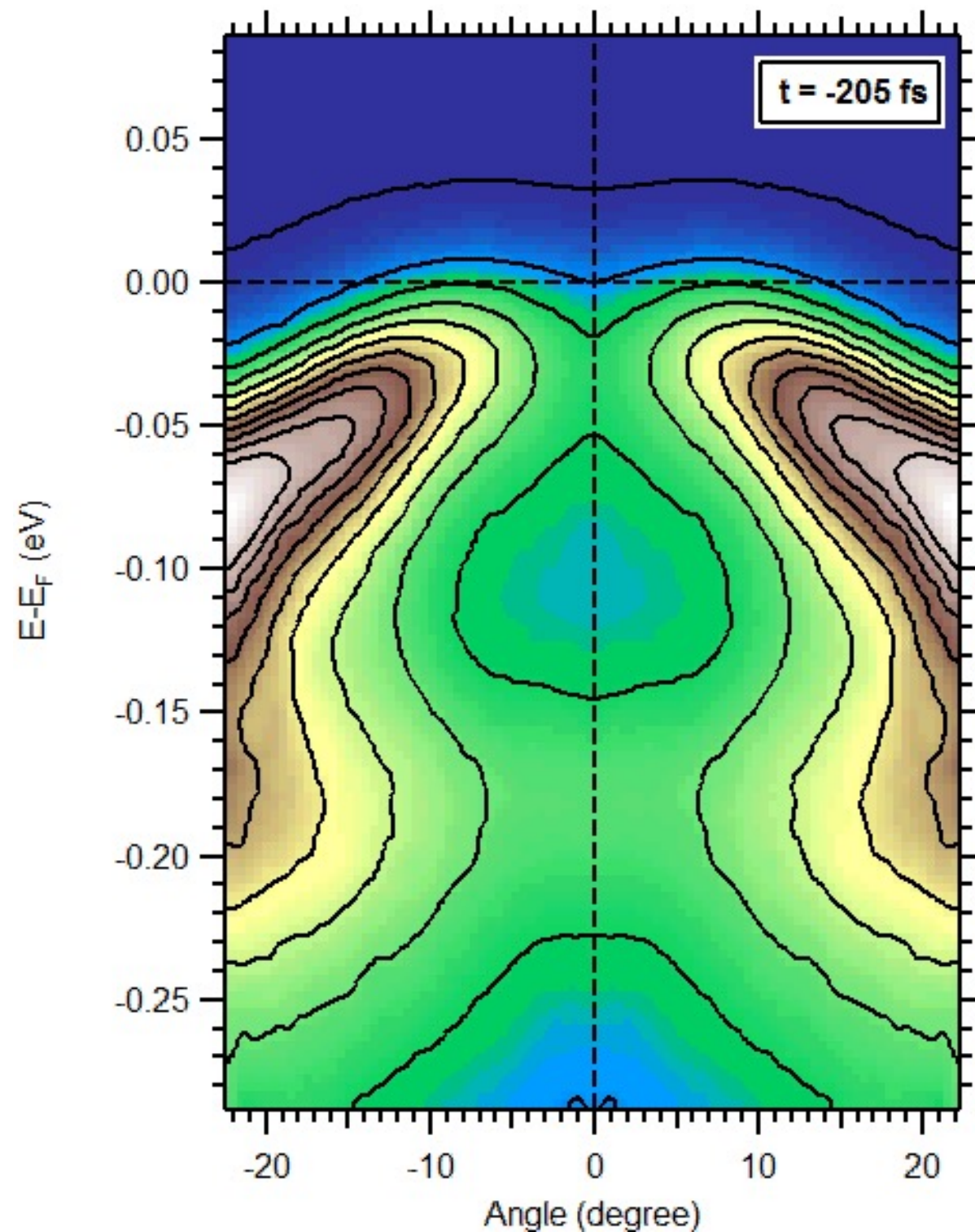
Slides courtesy of Shuolong Yang (U. Chicago)

Surface states in topological insulators

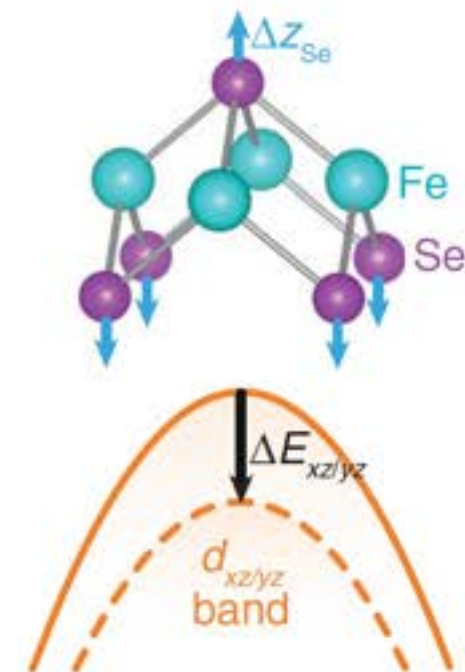
Topological insulator Bi_2Se_3



Coherent vibration of FeSe electronic bands



Perturbation launches coherent modes



What is Time-Resolved ARPES good for?

trARPES tool set

Resolving lifetimes

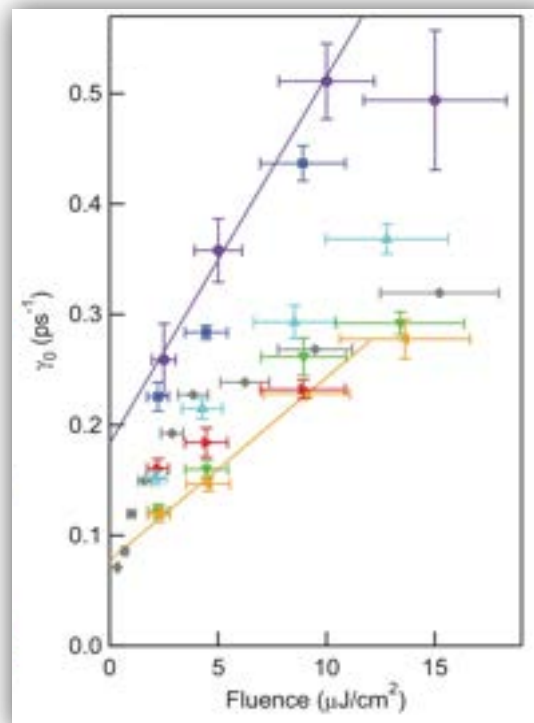
Launching collective excitations

Promoting electrons above E_F

Transiently modifying emergent properties



Cuprates [1]



Material physics problems

Complex scattering processes

Collective modes

Unoccupied band structure

Material control



[1] Smallwood *et al. Science* **336**, 1137 (2012) [2] Schmitt *et al. Science* **321**, 1649 (2008)

[3] Sobota *et al. Phys. Rev. Lett.* **111**, 136802 (2013) [4] Mahmood *et al. Nat. Phys.* Advance Online Publication (2016)

What is Time-Resolved ARPES good for?

trARPES tool set

Resolving lifetimes

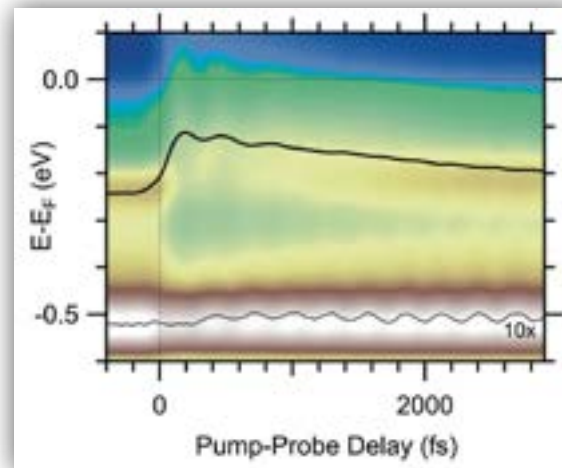
Launching collective excitations

Promoting electrons above E_F

Transiently modifying emergent properties



TbTe₃ [2]



Material physics problems

Complex scattering processes

Collective modes

Unoccupied band structure

Material control



[1] Smallwood *et al. Science* **336**, 1137 (2012) [2] Schmitt *et al. Science* **321**, 1649 (2008)

[3] Sobota *et al. Phys. Rev. Lett.* **111**, 136802 (2013) [4] Mahmood *et al. Nat. Phys.* Advance Online Publication (2016)

What is Time-Resolved ARPES good for?

trARPES tool set

Resolving lifetimes

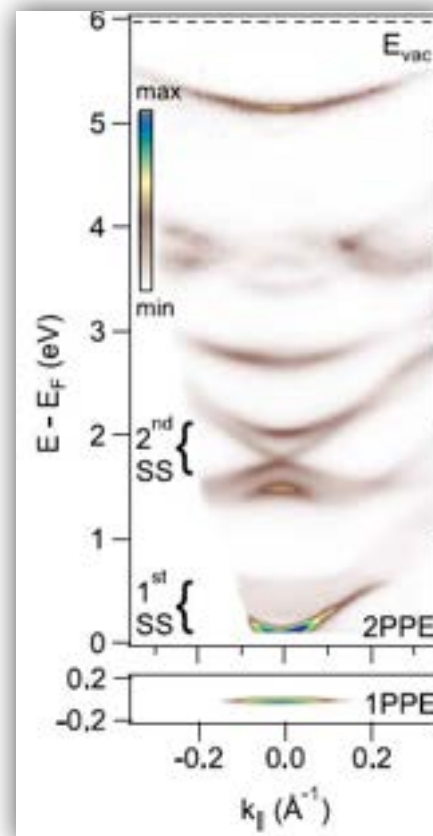
Launching collective excitations

Promoting electrons above E_F

Transiently modifying emergent properties



Bi_2Se_3 [3]



Material physics problems

Complex scattering processes

Collective modes

Unoccupied band structure

Material control



[1] Smallwood *et al.* *Science* **336**, 1137 (2012) [2] Schmitt *et al.* *Science* **321**, 1649 (2008)

[3] Sobota *et al.* *Phys. Rev. Lett.* **111**, 136802 (2013) [4] Mahmood *et al.* *Nat. Phys.* Advance Online Publication (2016)

What is Time-Resolved ARPES good for?

trARPES tool set

Resolving lifetimes

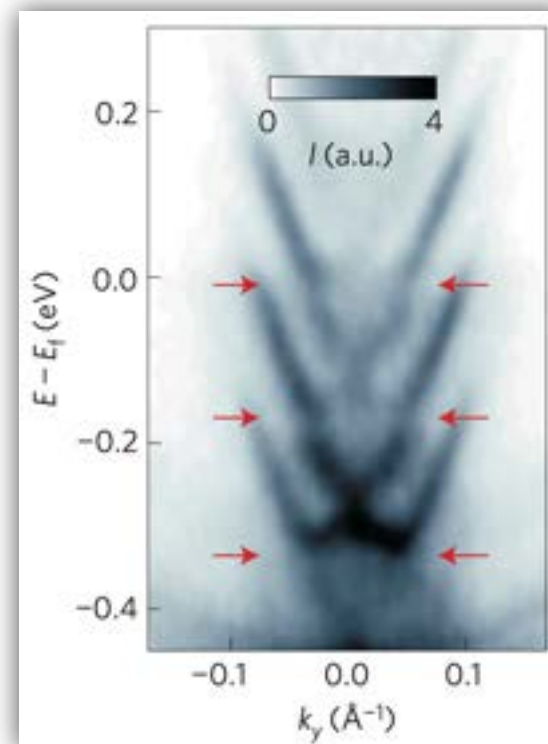
Launching collective excitations

Promoting electrons above E_F

Transiently modifying emergent properties



Bi_2Se_3 [4]



Material physics problems

Complex scattering processes

Collective modes

Unoccupied band structure

Material control



[1] Smallwood *et al.* *Science* **336**, 1137 (2012) [2] Schmitt *et al.* *Science* **321**, 1649 (2008)

[3] Sobota *et al.* *Phys. Rev. Lett.* **111**, 136802 (2013) [4] Mahmood *et al.* *Nat. Phys.* Advance Online Publication (2016)

TR-ARPES

some useful reviews

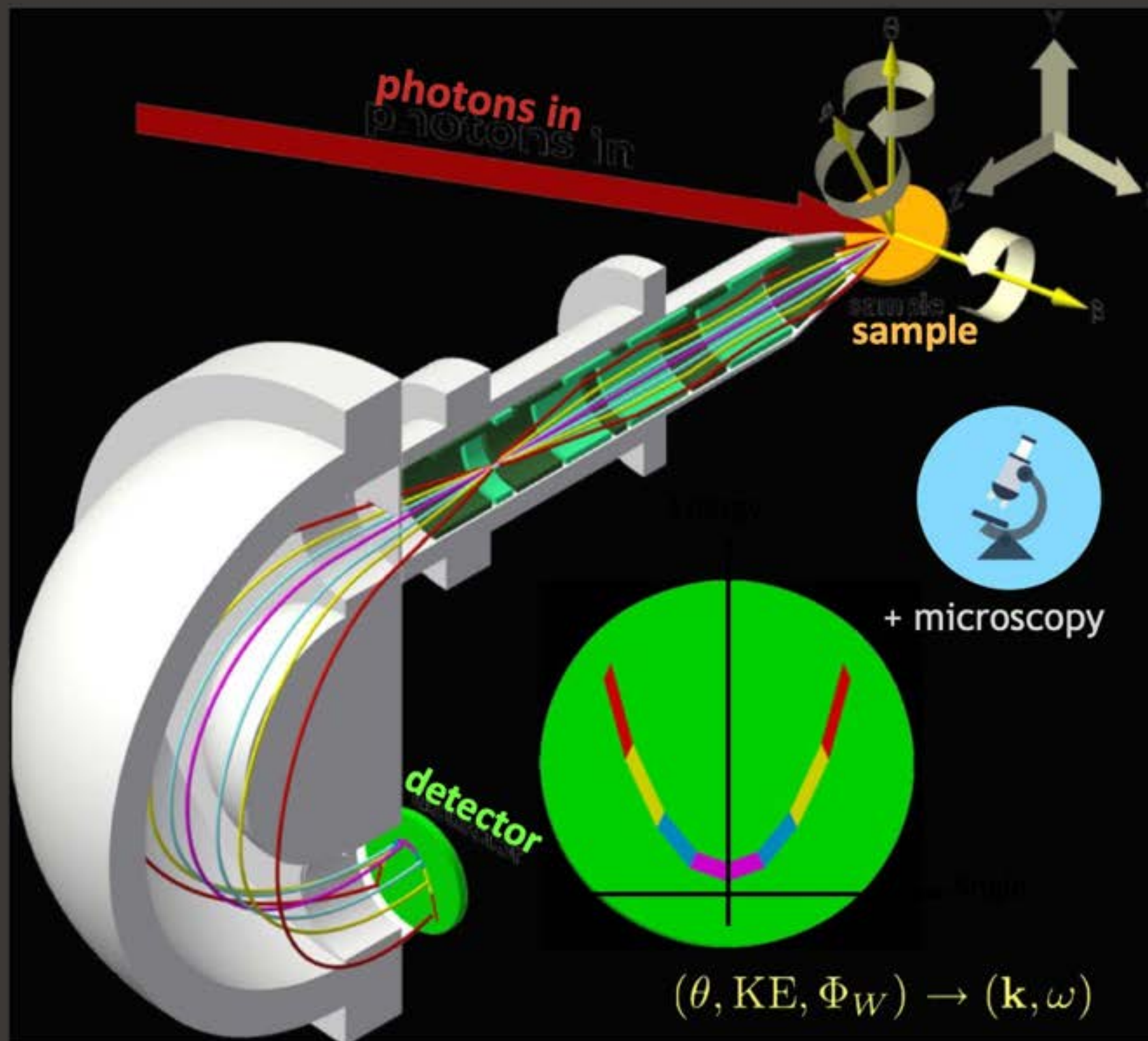
experiment

- S. Mathias *et al.*, J. of Phys.: Conf. Ser. **148**, 012042 (2009)
- U. Bovensiepen and P. S. Kirchmann, Laser Photonics Rev. **6**, 589 (2012)
- C. Giannetti *et al.*, Adv. in Phys. **65**, 58 (2016)
- C. L. Smallwood *et al.*, Europhys. Lett. **115**, 27001 (2016)

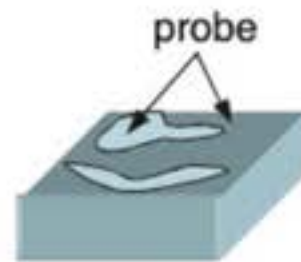
theory

- H. Aoki *et al.*, Rev. Mod. Phys. **86**, 779 (2014)
- A. F. Kemper *et al.*, Ann. Phys. **1600235** (2017)

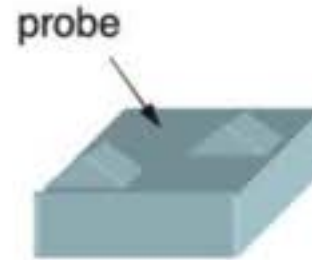
Angle-resolved photoelectron spectroscopy + something else



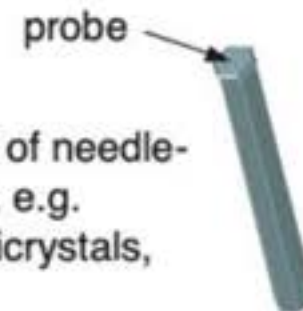
The case for going smaller



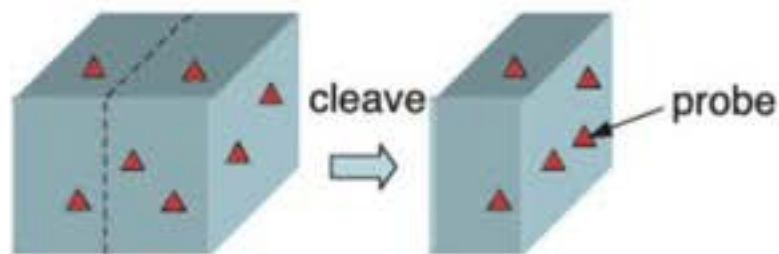
(a) phase separation
- doped Mott insulators
- magnetism



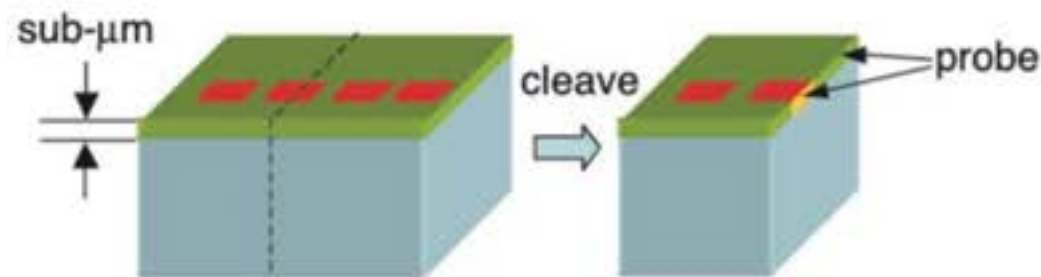
(b) isolating flat regions of
irregular cleaves



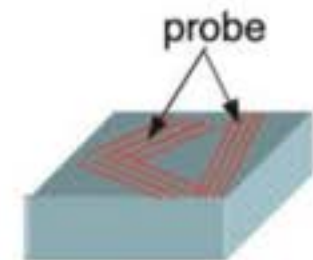
(c) 2-d plane of needle-
like samples, e.g.
 NbSe_3 , quasicrystals,
etc



(d) microcrystallites embedded
in a host material for cleavage

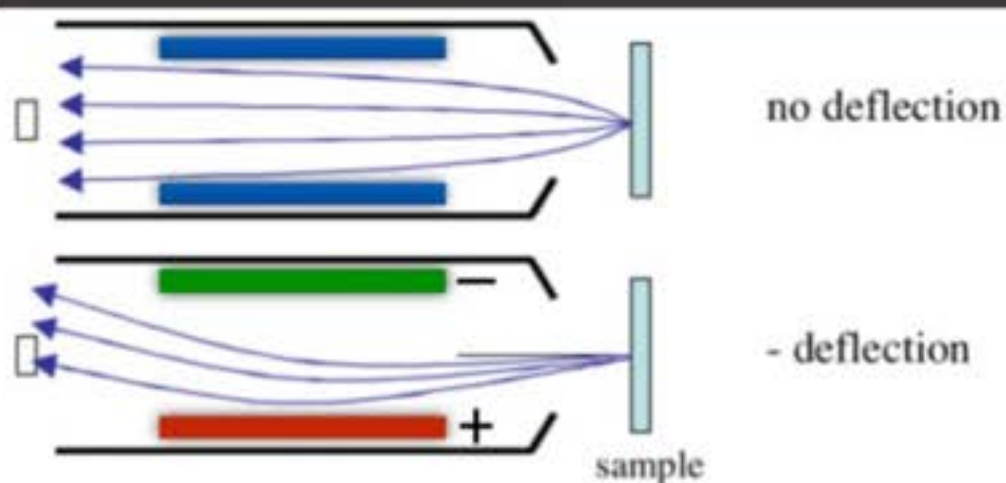
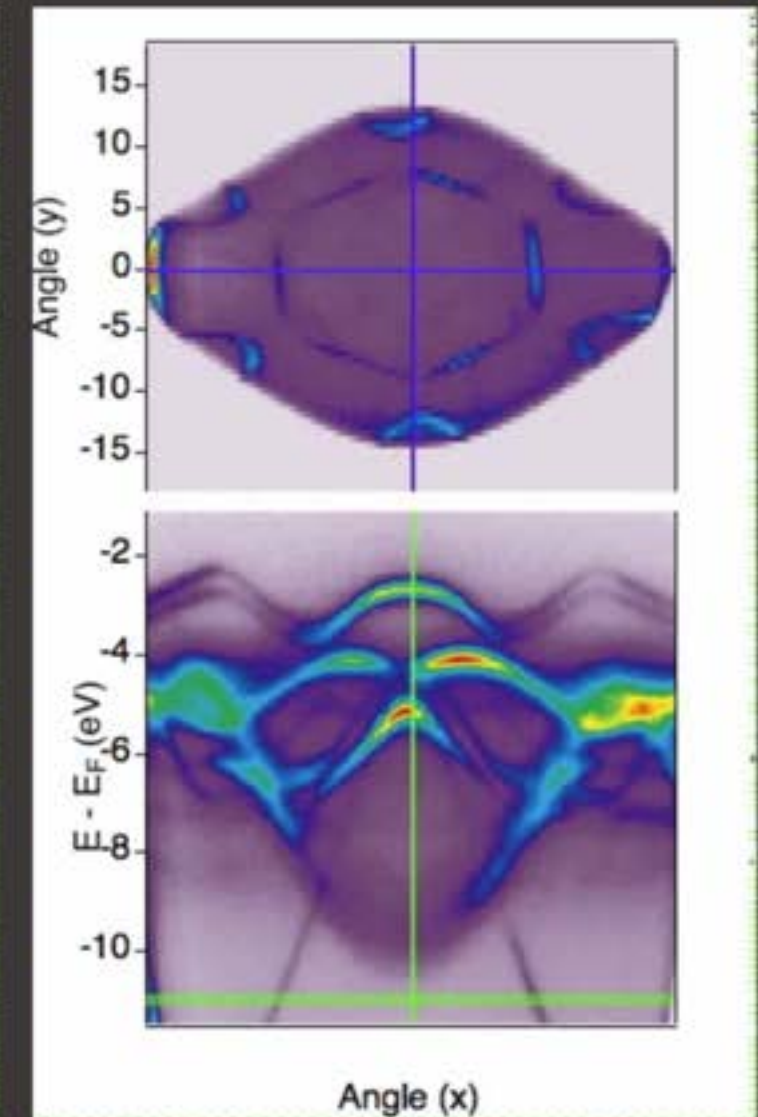
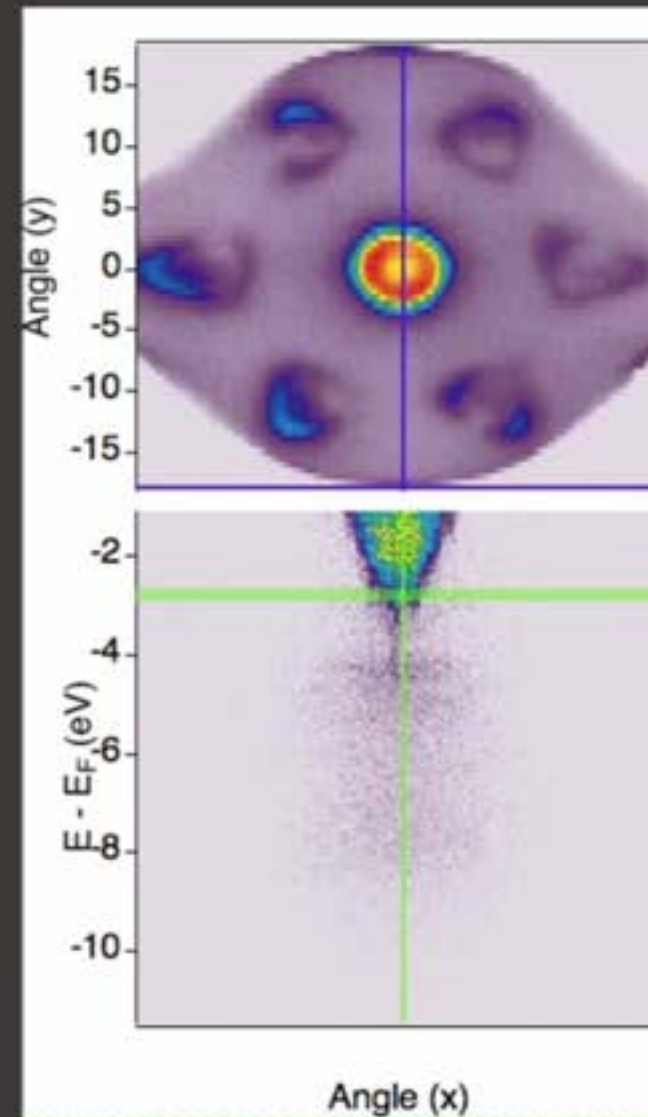
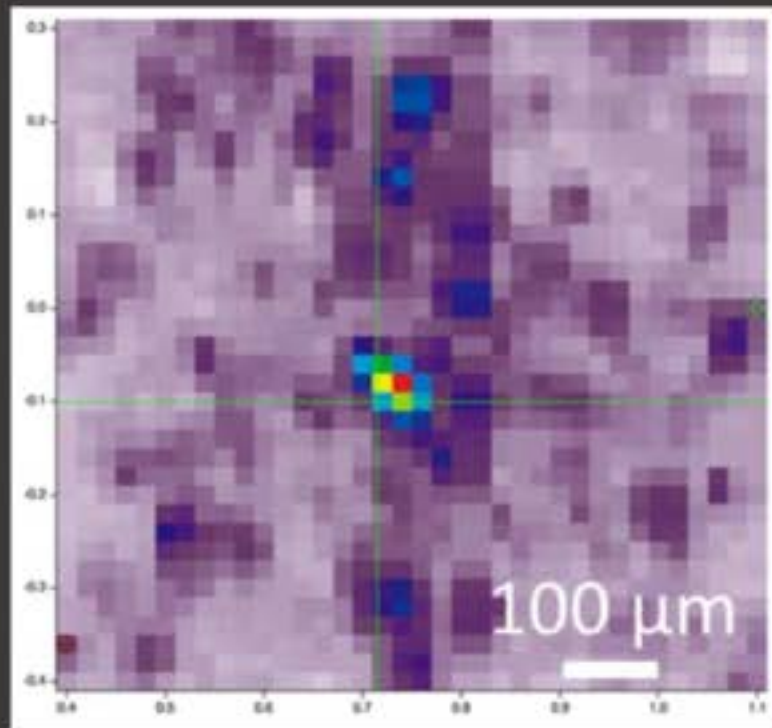
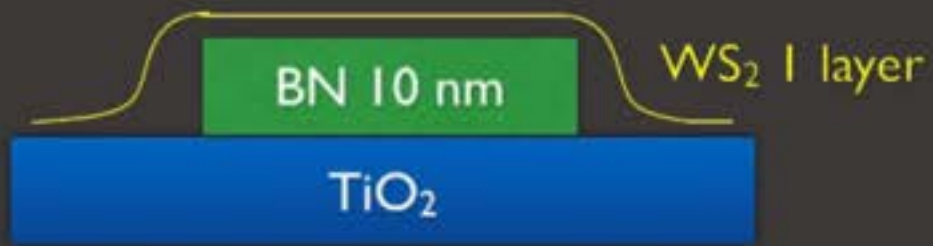


(e) thin films grown ex situ;
also quantum dots, other nano-
engineered devices



(f) isolating mixed phases
on epitaxial film surfaces

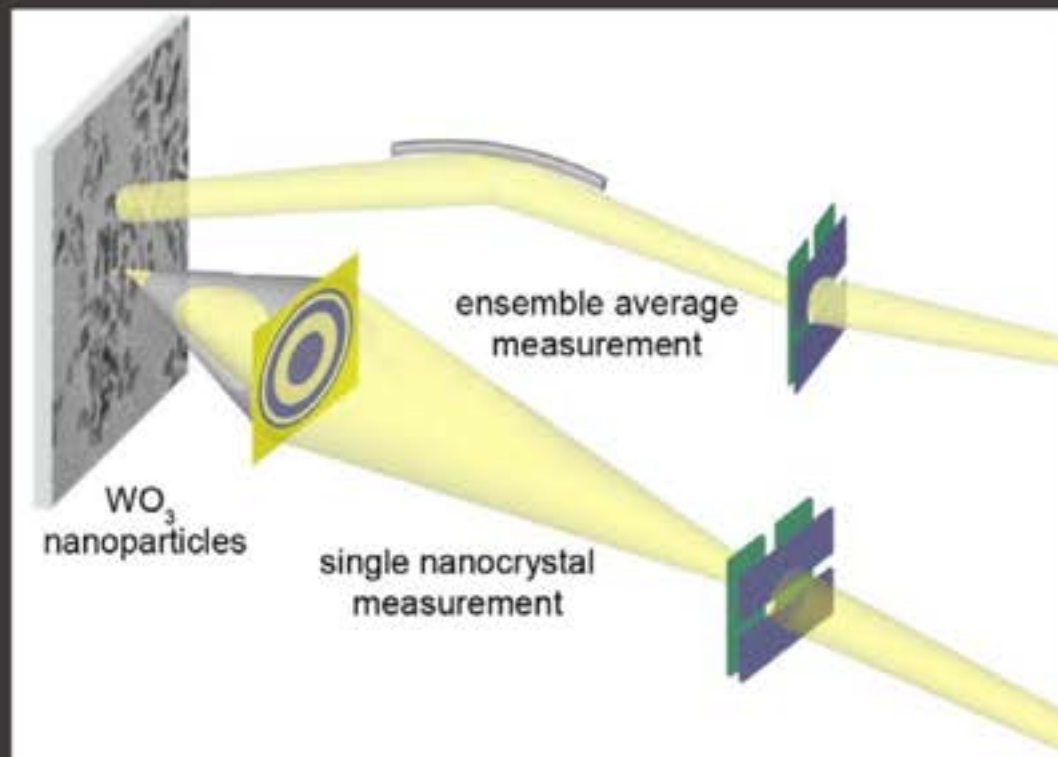
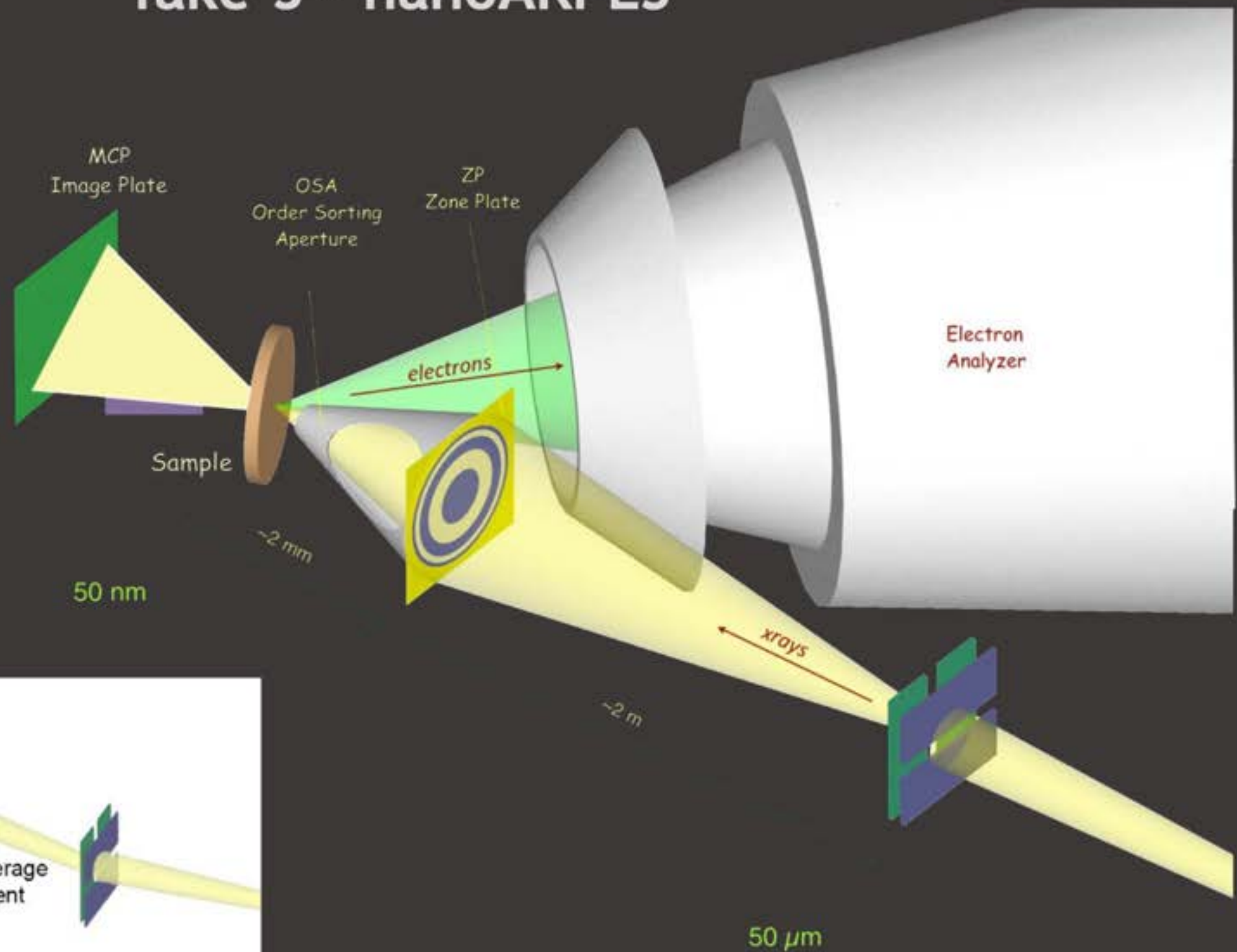
Take 1 – μ ARPES with deflectors



data: J. Katoch and S. Ulstrup

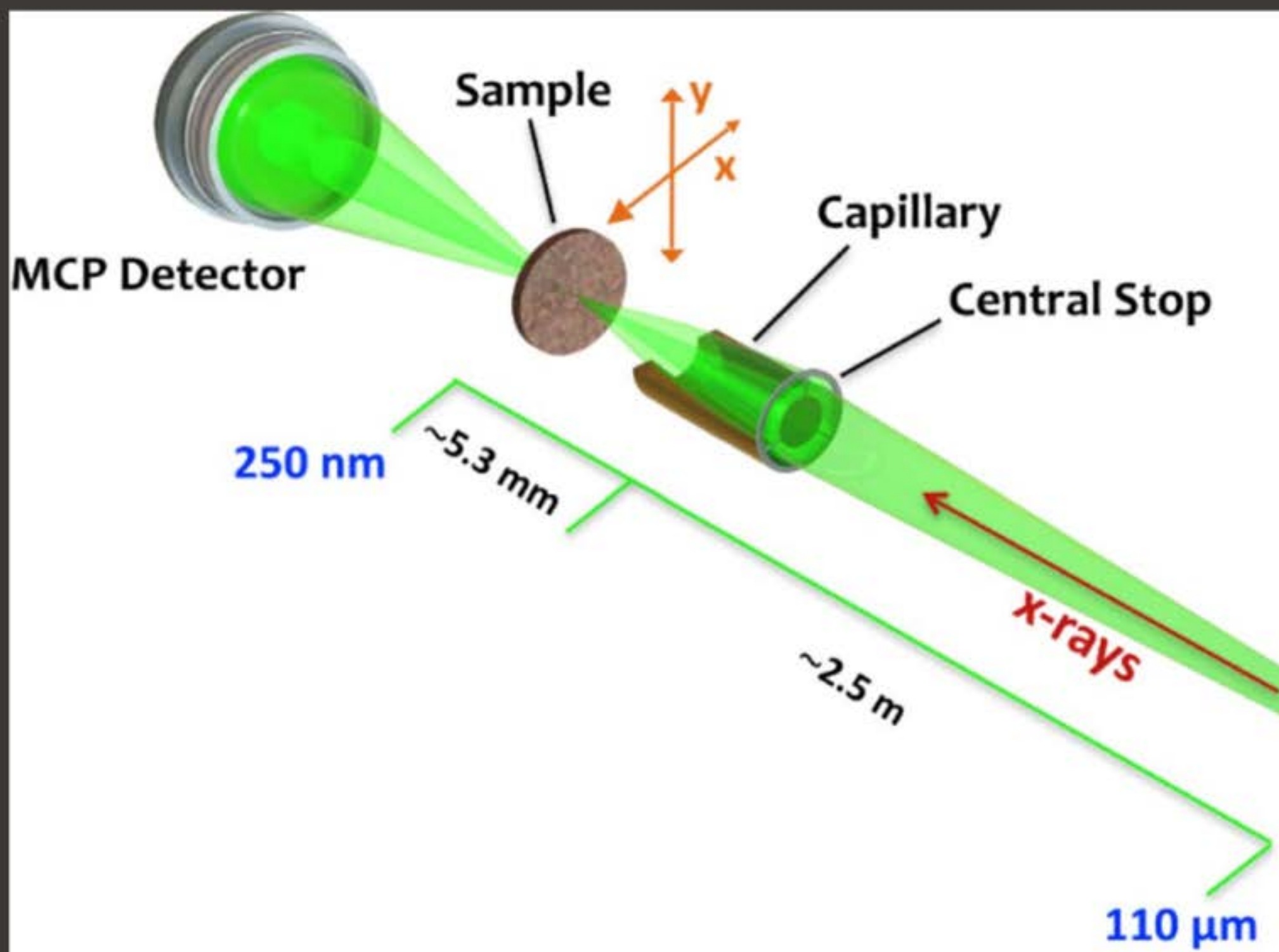
ackn. E. Rotenberg, A. Bostwick, R. Koch

Take 3 - nanoARPES



<120 nm spatial resolution
100 times slower than μARPES

(almost) nanoARPES with KB optics



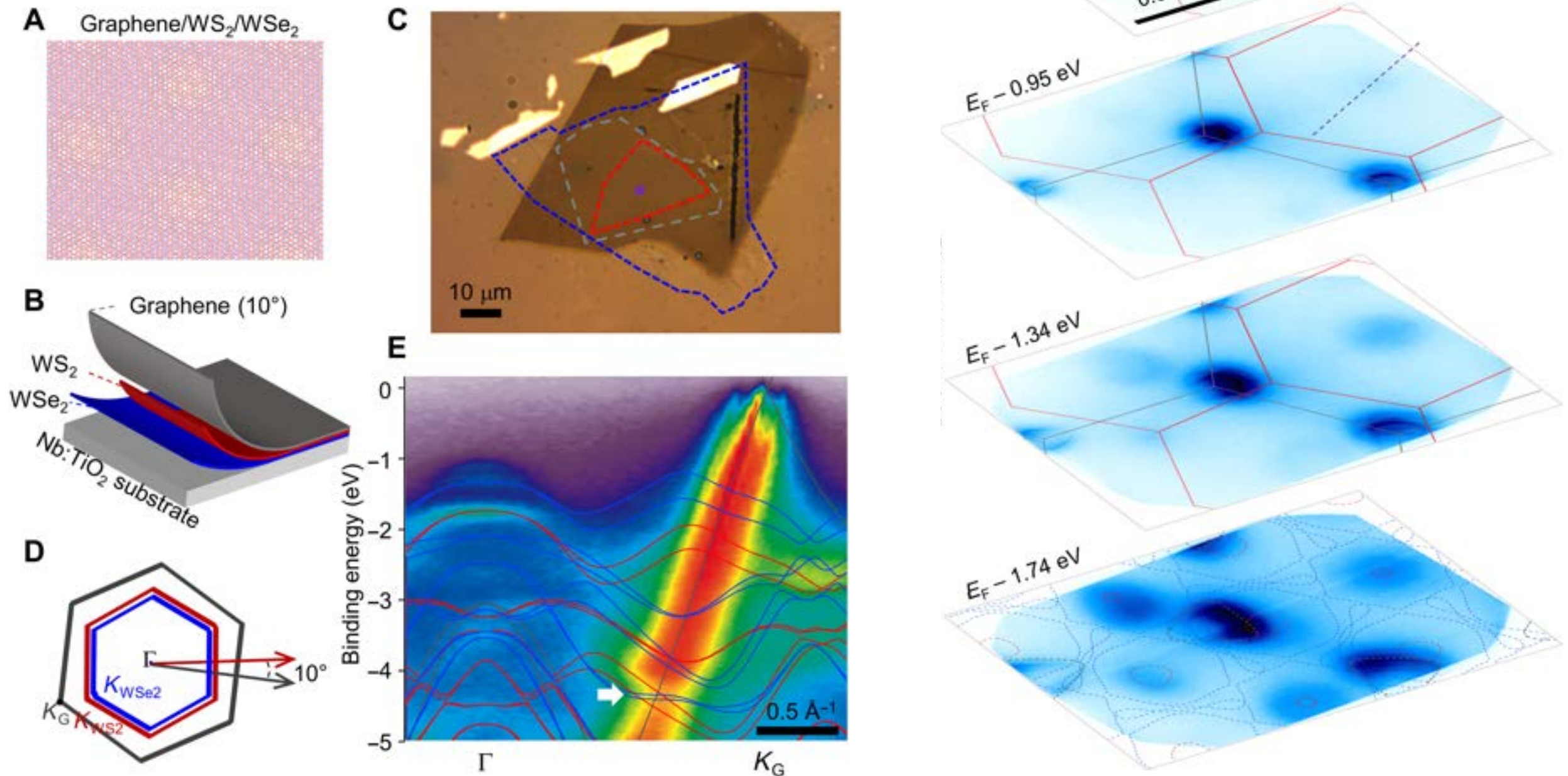
Spatially Resolved ARPES : on micro-structures

SCIENCE ADVANCES | RESEARCH ARTICLE

PHYSICAL SCIENCES

Strong interlayer interactions in bilayer and trilayer moiré superlattices

Saien Xie^{1,2,3*}, Brendan D. Faeth¹, Yanhao Tang⁴, Lizhong Li⁴, Eli Gerber⁴, Christopher T. Parzyck¹, Debanjan Chowdhury¹, Ya-Hui Zhang⁵, Christopher Jozwiak⁶, Aaron Bostwick⁶, Eli Rotenberg⁶, Eun-Ah Kim¹, Jie Shan^{1,3,4}, Kin Fai Mak^{1,3,4}, Kyle M. Shen^{1,3*}



Spatially Resolved ARPES : on micro-structures

